Manage IPsec policies

ONTAP 9.8 REST API reference

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Manage IPsec policies

Security IPsec policies endpoint overview

Overview

The following operations are supported:

- Collection Get: GET security/ipsec/policies
- Creation Post: POST security/ipsec/policies
- Instance Get: GET security/ipsec/policies/uuid
- Instance Patch: PATCH security/ipsec/policies/uuid
- Instance Delete: DELETE security/ipsec/policies/uuid

Retrieve IPsec policies

GET `/security/ipsec/policies`

Introduced In: 9.8

Retrieves the collection of IPsec policies.

Related ONTAP commands

- `security ipsec policy show`

Parameters

Name	Type	In	Required	Description
uuid	string	query	False	Filter by uuid
name	string	query	False	Filter by name
protocol	string	query	False	Filter by protocol
scope	string	query	False	Filter by scope
remote_endpoint.netmask	string	query	False	Filter by remote_endpoint.netmask
remote_endpoint.family	string	query	False	Filter by remote_endpoint.family

Name	Type	In	Required	Description
remote_endpoint.address	string	query	False	Filter by remote_endpoint.address
enabled	boolean	query	False	Filter by enabled
svm.uuid	string	query	False	Filter by svm.uuid
svm.name	string	query	False	Filter by svm.name
local_endpoint.netmask	string	query	False	Filter by local_endpoint.netmask
local_endpoint.family	string	query	False	Filter by local_endpoint.family
local_endpoint.address	string	query	False	Filter by local_endpoint.address
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
error	error	
num_records	integer	Number of records
records	array[records]	

Example response

```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  },
  "num_records": "1",
  "records": {
    "action": "bypass",
    "local_endpoint": {
      "address": "10.10.10.7",
      "family": "ipv4",
      "netmask": "24"
    },
    "protocol": "17",
    "remote_endpoint": {
      "address": "10.10.10.7",
      "family": "ipv4",
      "netmask": "24"
    },
    "scope": "svm",
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  }
}
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

local_endpoint

Local endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6

Name	Type	Description
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

remote_endpoint

Remote endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

_links

Name	Type	Description
self	href	

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

records

IPsec policy object.

Name	Type	Description
action	string	Action for the IPsec policy.

Name	Type	Description
enabled	boolean	Indicates whether or not the policy is enabled.
local_endpoint	local_endpoint	Local endpoint for the IPsec policy.
name	string	IPsec policy name.
protocol	string	Lower layer protocol to be covered by the IPsec policy.
remote_endpoint	remote_endpoint	Remote endpoint for the IPsec policy.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
secret_key	string	Pre-shared key for IKE negotiation.
svm	svm	SVM, applies only to SVM-scoped objects.
uuid	string	Unique identifier of the IPsec policy.

Create an IPsec policy

POST `/security/ipsec/policies`

Introduced In: 9.8

Creates an IPsec policy.

Related ONTAP commands

- `security ipsec policy create`

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
action	string	Action for the IPsec policy.
enabled	boolean	Indicates whether or not the policy is enabled.
local_endpoint	local_endpoint	Local endpoint for the IPsec policy.
name	string	IPsec policy name.
protocol	string	Lower layer protocol to be covered by the IPsec policy.
remote_endpoint	remote_endpoint	Remote endpoint for the IPsec policy.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
secret_key	string	Pre-shared key for IKE negotiation.
svm	svm	
uuid	string	Unique identifier of the IPsec policy.

Example request

```
{
  "action": "bypass",
  "local_endpoint": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "protocol": "17",
  "remote_endpoint": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "scope": "svm",
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```

Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
error	error	
num_records	integer	Number of records
records	array[records]	

Example response

```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  },
  "num_records": "1",
  "records": {
    "action": "bypass",
    "local_endpoint": {
      "address": "10.10.10.7",
      "family": "ipv4",
      "netmask": "24"
    },
    "protocol": "17",
    "remote_endpoint": {
      "address": "10.10.10.7",
      "family": "ipv4",
      "netmask": "24"
    },
    "scope": "svm",
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  }
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
66257099	Only one protocol can be specified.
66257100	Only one local port can be specified.
66257101	Only one remote port can be specified.
66257103	IPsec policy with same name already exists in this SVM.
66257107	The specified pre-shared key is not a valid hexadecimal string.
66257109	The specified pre-shared key is not a valid Base64 encoded binary string.
66257110	Failed to a create policy sequencing value.
66257111	The IPsec policy with action ESP TRANSPORT provides packet protection and requires a secret key for authentication.
66257112	The IPsec policy with the action specified does not provide packet protection and the authentication method provided for the policy will be ignored.
66257113	Only one local IP subnet can be specified.
66257114	Only one remote IP subnet can be specified.
66257115	Port ranges containing more than one port are not supported.
66257117	IPsec is not supported on the SVM specified in the policy, IPsec is supported on data SVMs only.
66257120	The subnet selector must be a host address (An IPv4 address with a 32-bit netmask or an IPv6 address with a 128-bit netmask).
66257121	The maximum limit of IPsec Policies has reached for the specified SVM.
66257125	The local_endpoint.address must be specified with local_endpoint.netmask.
66257126	The remote_endpoint.address must be specified with remote_endpoint.netmask.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

local_endpoint

Local endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

remote_endpoint

Remote endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

svm

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

ipsec_policy

IPsec policy object.

Name	Type	Description
action	string	Action for the IPsec policy.
enabled	boolean	Indicates whether or not the policy is enabled.
local_endpoint	local_endpoint	Local endpoint for the IPsec policy.
name	string	IPsec policy name.
protocol	string	Lower layer protocol to be covered by the IPsec policy.
remote_endpoint	remote_endpoint	Remote endpoint for the IPsec policy.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
secret_key	string	Pre-shared key for IKE negotiation.
svm	svm	
uuid	string	Unique identifier of the IPsec policy.

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

records

IPsec policy object.

Name	Type	Description
action	string	Action for the IPsec policy.
enabled	boolean	Indicates whether or not the policy is enabled.
local_endpoint	local_endpoint	Local endpoint for the IPsec policy.
name	string	IPsec policy name.

Name	Type	Description
protocol	string	Lower layer protocol to be covered by the IPsec policy.
remote_endpoint	remote_endpoint	Remote endpoint for the IPsec policy.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
secret_key	string	Pre-shared key for IKE negotiation.
svm	svm	SVM, applies only to SVM-scoped objects.
uuid	string	Unique identifier of the IPsec policy.

Delete an IPsec policy

DELETE `/security/ipsec/policies/{uuid}`

Introduced In: 9.8

Deletes a specific IPsec policy.

Related ONTAP commands

- `security ipsec policy delete`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	IPsec policy UUID

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
66257096	Internal error. Failed to purge connections associated with the IPsec policy.
66257116	IPsec policy with the specified UUID was not found.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve an IPsec policy

GET `/security/ipsec/policies/{uuid}`

Introduced In: 9.8

Retrieves a specific IPsec policy.

Related ONTAP commands

- `security ipsec policy show`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	IPsec policy UUID
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
action	string	Action for the IPsec policy.
enabled	boolean	Indicates whether or not the policy is enabled.
local_endpoint	local_endpoint	Local endpoint for the IPsec policy.
name	string	IPsec policy name.
protocol	string	Lower layer protocol to be covered by the IPsec policy.
remote_endpoint	remote_endpoint	Remote endpoint for the IPsec policy.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
secret_key	string	Pre-shared key for IKE negotiation.
svm	svm	
uuid	string	Unique identifier of the IPsec policy.

Example response

```
{
  "action": "bypass",
  "local_endpoint": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "protocol": "17",
  "remote_endpoint": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "scope": "svm",
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
66257116	IPsec policy with the specified UUID was not found.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

local_endpoint

Local endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

remote_endpoint

Remote endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

svm

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update an IPsec policy

PATCH `/security/ipsec/policies/{uuid}`

Introduced In: 9.8

Updates a specific IPsec policy.

Related ONTAP commands

- `security ipsec policy modify`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	IPsec policy UUID

Request Body

Name	Type	Description
action	string	Action for the IPsec policy.
enabled	boolean	Indicates whether or not the policy is enabled.
local_endpoint	local_endpoint	Local endpoint for the IPsec policy.
name	string	IPsec policy name.
protocol	string	Lower layer protocol to be covered by the IPsec policy.
remote_endpoint	remote_endpoint	Remote endpoint for the IPsec policy.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
secret_key	string	Pre-shared key for IKE negotiation.
svm	svm	
uuid	string	Unique identifier of the IPsec policy.

Example request

```
{
  "action": "bypass",
  "local_endpoint": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "protocol": "17",
  "remote_endpoint": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "scope": "svm",
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
66257097	Internal error. Failed to update the IPsec policy.
66257099	Only one protocol can be specified.

Error Code	Description
66257100	Only one local port can be specified.
66257101	Only one remote port can be specified.
66257110	Failed to create a policy sequencing value.
66257113	Only one local IP subnet can be specified.
66257114	Only one remote IP subnet can be specified.
66257115	Port ranges containing more than one port are not supported.
66257116	IPsec policy with the specified UUID was not found.
66257120	The subnet selector must be a host address (An IPv4 address with a 32-bit netmask or an IPv6 address with a 128-bit netmask).

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

local_endpoint

Local endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

remote_endpoint

Remote endpoint for the IPsec policy.

Name	Type	Description
address	string	IPv4 or IPv6 address
family	string	IPv4 or IPv6
netmask	string	Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, you must set the netmask length. The default value is 64. Output is always netmask length.

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

svm

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	The name of the SVM.
uuid	string	The unique identifier of the SVM.

ipsec_policy

IPsec policy object.

Name	Type	Description
action	string	Action for the IPsec policy.
enabled	boolean	Indicates whether or not the policy is enabled.
local_endpoint	local_endpoint	Local endpoint for the IPsec policy.
name	string	IPsec policy name.
protocol	string	Lower layer protocol to be covered by the IPsec policy.
remote_endpoint	remote_endpoint	Remote endpoint for the IPsec policy.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
secret_key	string	Pre-shared key for IKE negotiation.
svm	svm	
uuid	string	Unique identifier of the IPsec policy.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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